

Abhikalp Unakal

Software Engineer

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Professional Experience

Meta, Software Engineer

2021 – 2026 | Redmond | Vancouver

- Audio Research at Meta Reality Labs - Shipped "Conversation Focus" on Ray Ban Meta glasses
<https://www.yahoo.com/entertainment/articles/meta-connect-2025-showcases-three-151420069.html>
- Architected and built the metrics and telemetry systems for Ray-Ban Meta glasses and research prototypes that enabled real time live tuning of DSP parameters and transitioned the team from monthly to continuous hourly algorithmic validations using C++, C#, and Python.
- Led a team to engineer and architect a petabyte scale multimodal data collection warehouse integrating custom networking, hardware, multiple sensor types, motion capture, metrics, and telemetry into deep learning annotation pipelines using C++, C#, and Python.
- Built a C++/ImGui audio graph visualizer that auto generates DSP pipeline diagrams and graph specifications - similar to Unreal Blueprints but for embedded systems, enabling proactive flagging of code changes that break the signal chain.
- Upgraded Meta's telemetry system (unidash) from 1D to N dimensional plotting using Python and Hack, enabling frequency response curves as one of the main use cases for 80,000 engineers.
- Coordinated and built necessary infrastructure via fbpkgs + custom licensing layer to enable Matlab code to run in the cloud - which enabled feature parity checks across research and product teams.
- Developed a C# based hardware validation rig integrated with Unity to track head pose (azimuth/elevation) using a fine-tuned YOLOv3 model. The system automates video frame alignment for photogrammetry, eliminating the need for manual participant positioning by Research Assistants.

Roland Corporation US, Software Engineering Intern

2020 – 2021 | Redmond

- Led UX engineering for the device editor as part of the Roland Cloud Editor Team to create a Virtual Tone editor that lets JUPITER-X users manage their sound profiles. I was responsible for the UX design and built a programmatic auto layout engine for the Roland - JUPITER-X Editor | Editor/Librarian. https://www.roland.com/us/products/rc-jupiter-x_editor/

Danske Bank, Software Engineer

2016 – 2019 | Bengaluru | Copenhagen

- Designed and deployed customized pipelines for efficient data model deployment framework with a custom Heartbeat detection layer on a Hadoop cluster with 50 Hadoop boxes and 250 edge nodes reduced deployments from 3 months to 1 week.

Education

M.S. Computer Science, DigiPen Institute of Technology

Redmond, USA

B.E. Computer Science, PES Institute of Technology

Bengaluru, India

Skills

C++ • C • Python • Go • Javascript • C# •
Audio DSP • FastAPI • Torch • Numpy • OpenGL •
Nodejs • React JS • Bash • Unity • Godot •
Game Engines • Game Design • PostgreSQL •
Logic Pro • Music Production • 3D math • ML •
Langchain • RAG • Distributed Systems • Graphs

Achievements

Shipped Conversation Focus, Meta Rayban Smartglasses

Shipped Jupiter X editor, Roland Cloud

San Jose State University, Exchange Student with 89%

All India Student Design Competition,
Runner Up - event organized by NID

Danske Bank, Runner up - case study competition

Projects

MoorerReverb Research Paper Implementation, Real Time Reverb Engine in JUCE

- Written from scratch using c++ and JUCE this is a VST plugin that can work in any DAW. This is a research paper implementation of James A Moorer's Research Paper. <https://github.com/auxeon/MoorerReverb>

Katana Utility AI, Custom Utility AI for football 2D

- Football AI in C# using unity that employs utility based behavior behavior trees. One of the most fun projects I worked on. <https://www.youtube.com/watch?v=rBCwW9KNq9A>

Atom, Game Engine Built from Scratch using c++

- I was the Tech lead + producer on this project and managed 3 other engineers. This game engine had advanced Event systems, ECS engine and 3D audio. <https://www.youtube.com/watch?v=qDjw00-M0QI>

Space Jump, Original Platformer spatial puzzle game with 15+ levels

- I was the lead game designer for this unique platform puzzle with a novel teleporation mechanism with numerous powerups and boss battles with 15+ levels. which was designed and built on a custom in house engine built on the custom game engine Atom. <https://github.com/auxeon/Atom>

Ctest, Fast simple header only test framework for c, c++

- A simple Header only test framework in c99 with the following features with all the features of the heavyweight google test - written using c99 with integrations with the buck2 build system. <https://github.com/auxeon/ctest>